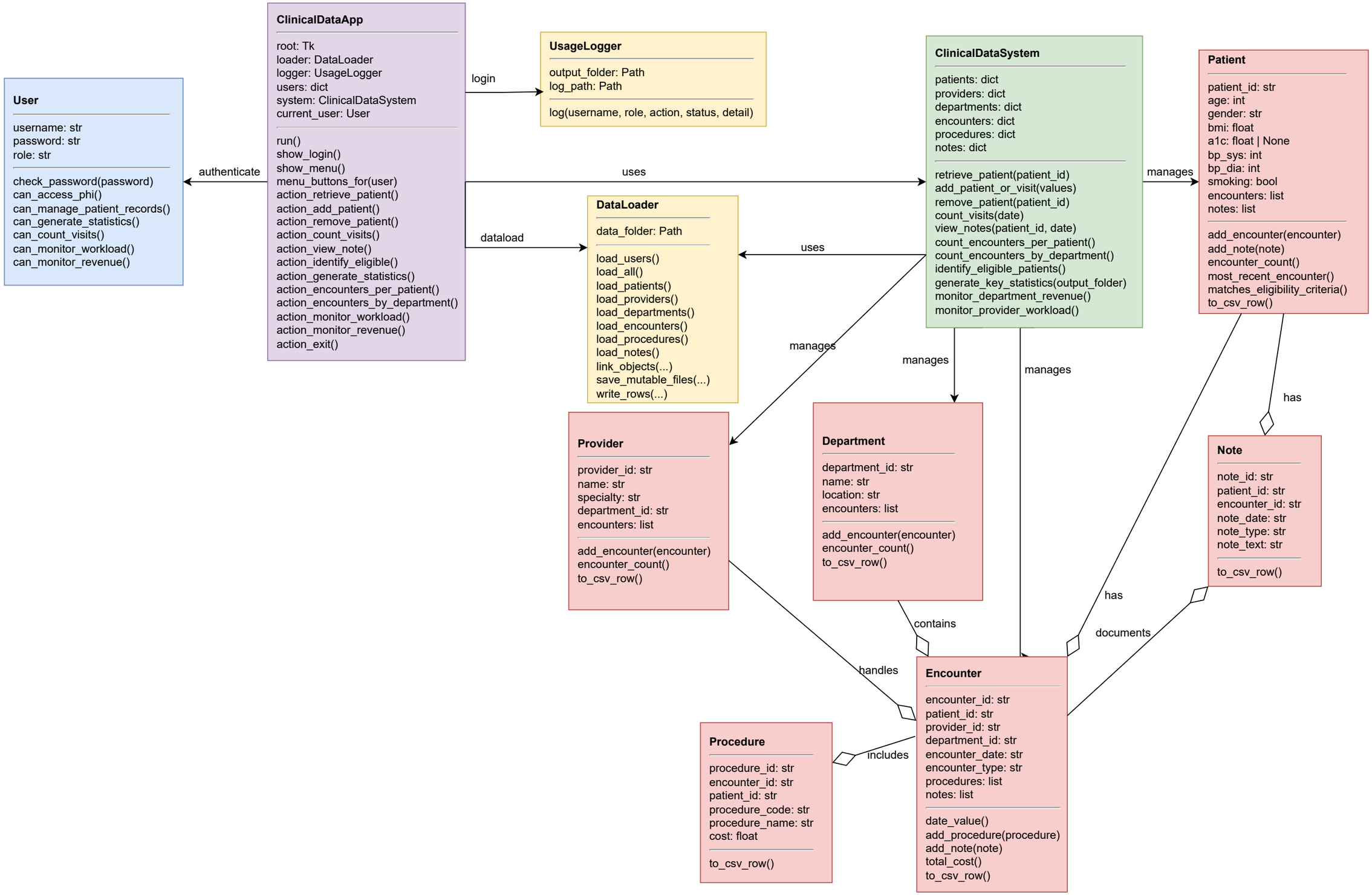
User
username: str
password: str
role: str
check_password(password)
can_access_phi()
can_manage_patient_records()
can_generate_statistics()
can_count_visits()
can_monitor_workload()
can_monitor_revenue()
authenticate
ClinicalDataApp
root: Tk
loader: DataLoader
logger: UsageLogger
users: dict
system: ClinicalDataSystem
current_user: User
run()
show_login()
show_menu()
menu_buttons_for(user)
action_retrieve_patient()
action_add_patient()
action_remove_patient()
action_count_visits()
action_view_note()
action_identify_eligible()
action_generate_statistics()
action_encounters_per_patient()
action_encounters_by_department()
action_monitor_workload()
action_monitor_revenue()
action_exit()
login
UsageLogger
output_folder: Path
log_path: Path
log(username, role, action, status, detail)
uses
dataload
DataLoader
data_folder: Path
load_users()
load_all()
load_patients()
load_providers()
load_departments()
load_encounters()
load_procedures()
load_notes()
link_objects(...)
save_mutable_files(...)
write_rows(...)
uses
ClinicalDataSystem
patients: dict
providers: dict
departments: dict
encounters: dict
procedures: dict
notes: dict
retrieve_patient(patient_id)
add_patient_or_visit(values)
remove_patient(patient_id)
count_visits(date)
view_notes(patient_id, date)
count_encounters_per_patient()
count_encounters_by_department()
identify_eligible_patients()
generate_key_statistics(output_folder)
monitor_department_revenue()
monitor_provider_workload()
manages
Patient
patient_id: str
age: int
gender: str
bmi: float
a1c: float | None
bp_sys: int
bp_dia: int
smoking: bool
encounters: list
notes: list
add_encounter(encounter)
add_note(note)
encounter_count()
most_recent_encounter()
matches_eligibility_criteria()
to_csv_row()
manages
manages
manages
has
Provider
provider_id: str
name: str
specialty: str
department_id: str
encounters: list
add_encounter(encounter)
encounter_count()
to_csv_row()
Department
department_id: str
name: str
location: str
encounters: list
add_encounter(encounter)
encounter_count()
to_csv_row()
Note
note_id: str
patient_id: str
encounter_id: str
note_date: str
note_type: str
note_text: str
to_csv_row()
contains
has
documents
handles
Encounter
encounter_id: str
patient_id: str
provider_id: str
department_id: str
encounter_date: str
encounter_type: str
procedures: list
notes: list
date_value()
add_procedure(procedure)
add_note(note)
total_cost()
to_csv_row()
includes
Procedure
procedure_id: str
encounter_id: str
patient_id: str
procedure_code: str
procedure_name: str
cost: float
to_csv_row()